

Advanced Data Management
2021/2022
February 11, 2022

Q1. (up to 10 lines) [**only for COMPUTER ENGINEERING STUDENTS**]

Discuss the main principles for large scale data management.

Q2. (up to 10 lines)

Describe and compare the column-family and the document data models.

Q3. (up to 10 lines)

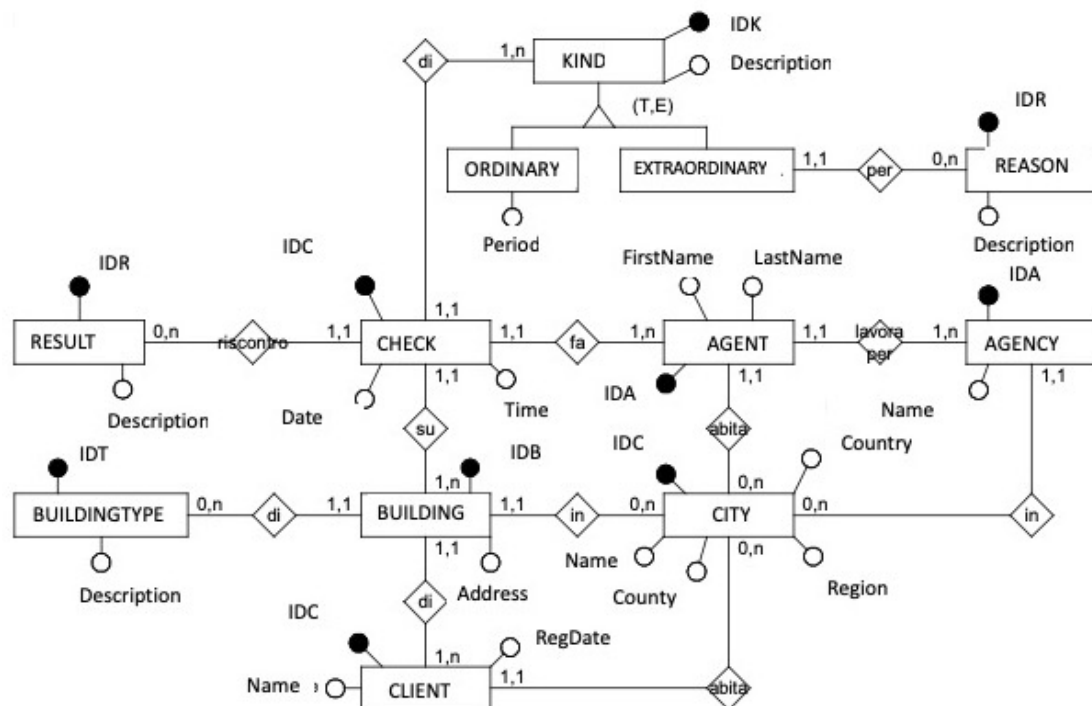
Compare column-family and graph data stores with respect to typically supported data distribution (partitioning, replication) facilities.

Q4. (up to 10 lines) [**only for COMPUTER SCIENCE STUDENTS**]

Discuss the differences between SELECT and CONSTRUCT SPARQL query forms.

Exercise 1 [NoSQL aggregate-oriented data stores]

Consider the following entity relationship schema:



The workload is the following:

- **Query 1:** Determine the client information together with the buildings she owns, their type and the city in which they are located.
- **Query 2:** Determine for each agency the checks made by agents in the agency, together with the kind and the reason for the check, the address of the checked building and the check result.
- **Query 3:** Determine for each reason of extraordinary check, the address of the checked building and the agent.

Based on the previous ER schema and workload:

1. Design, using the aggregate-oriented logical design methodology, the aggregates needed for answering the given workload queries; show the output of each step of the methodology.
2. Formulate Query 1 over the MongoDB document collection corresponding to the obtained aggregates.

Exercise 2 [NoSQL graph-based data stores]

Consider the following relational schema for car rentals (underlined attributes denote the primary key, superscripts denote referenced relation for foreign keys):

CUSTOMER(CustId, Name, Surname, Address, BirthDate, BirthPlace)

CAMPING(Name, Locality, Address, Category)

FACILITIES(Camping^{CAMPING}, Facility, Number)

PITCH(Camping^{CAMPING}, NumP, ByCar, HasElectricity, Sector)

VISIT(CustId^{CUSTOMER}, StartDate, EndDate, Camping^{PITCH}, NumP^{PITCH}, UseElectricity)

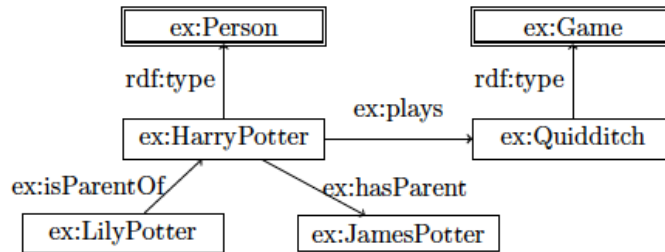
ACCOMPANYING(CustId^{VISIT}, StartDate^{VISIT}, AccompanyingNbr, Name, Surname, BirthDate, BirthPlace, AccId)

EQUIPMENT(CustId^{VISIT}, StartDate^{VISIT}, EquipmentNbr, Type, Description, Plate, Descr, LicencePlate₀)

1. Present the most suitable logical schema in a graph-based data store, like Neo4j (in the form of graph pattern or of a sample instance) for the domain described by the previous relational schema.
2. Formulate in Neo4j the query to retrieve the pairs of customers born on the same date that visited the same pitch.

Exercise 3 [Semantic data stores] [only for COMPUTER SCIENCE STUDENTS]

Consider the following RDF graph (ex:Person is a class, ex:HarryPotter is an instance):



- 1) Write the following RDF/RDFS statements in Turtle syntax:
 - a) James Potter and Lily Potter are persons.
 - b) A Person can be a Muggle (a person who cannot do magic) or a wizard (a person who can do magic).
 - c) ex:plays can take a person as subject and a game as an object.
- 2) Write the following OWL statements in XML syntax or turtle syntax:
 - a) James Potter is different from Lily Potter.
 - b) If a person A has a parent B, it always means that person B is the parent of A.
 - c) To be a Quidditch player, one has to play the game Quidditch.
 - d) A person can be a Muggle or a wizard, but nothing else.
- 3) Write the SPARQL query to determine all the wizards that play Quidditch OR are parent of a Muggle.